

EXPERIMENT 4

TITLE

Depth First Search (DFS)

AIM

To write a Python program to implement the **Depth First Search (DFS)** algorithm.

OBJECTIVE

- To understand Depth First Search.
- To traverse a graph using recursion.
- To visit every node exactly once.
- To demonstrate depth-wise traversal.

ALGORITHM

Step 1: Create the graph.

Step 2: Mark all nodes as unvisited.

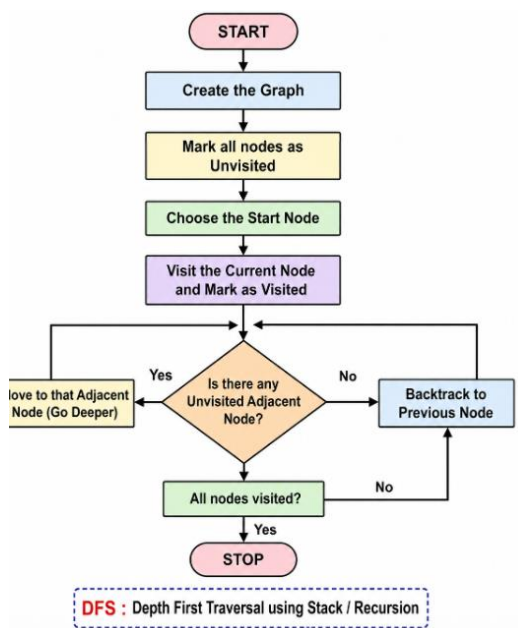
Step 3: Start from the source node.

Step 4: Visit the current node.

Step 5: Visit each unvisited adjacent node recursively.

Step 6: Continue until all nodes are visited.

FLOWCHART



PYTHON CODE

```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

visited = set()

def dfs(node):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)
        for neighbour in graph[node]:
            dfs(neighbour)

dfs('A')
```

SAMPLE OUTPUT

A B D E C F

CONCLUSION

The Depth First Search algorithm was implemented successfully. It traverses the graph by exploring each branch completely before backtracking.